Whose Contribution Counts?

AI Mediation and Collective Authorship in Digital Improvisation

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Exploratory proposal for discussion with Dr. Rebecca Caines
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Intended Fall 2027 entry

October 13, 2026

The research setting and partnerships will be developed in consultation with the prospective supervisor.

1 Problem and cultural stakes

In collaborative creative tools, a generative model does more than produce content: it *reinterprets* what people contribute. Someone types a phrase, and a moment later it has become a sound or a moving form in a shared scene. The group then has to decide what that transformation means, and whether it still belongs to the person who offered it.

This is a communication problem. Hancock et al. (2020) define AI-mediated communication as "interpersonal communication in which an intelligent agent operates on behalf of a communicator by modifying, augmenting, or generating messages to accomplish communication goals," with examples such as smart replies and auto-completion. In a shared creative work, the "message" is an artistic offer, its audience is the rest of the group, and the AI's modification becomes part of a collective artwork. When AI translates contributions in this way, its outputs may become more visible or influential than the contributions that prompted them, a possibility to investigate rather than an established effect. This raises a cultural question about whose interpretations gain authority, whose contributions remain recognisable, and how disagreement is negotiated. Creative platforms increasingly record who supplied what; this project asks whether such records settle questions of credit.

What is new is the object of study. The closest studies (Table 1) mostly position AI as a partner, performer or group resource that people create *with*. This project studies AI as a *medium* that people create *through*, a translator between collaborators whose reinterpretations they must accept, revise or refuse, and asks what that does to recognition and authorship over repeated collective work.

Improvisation is a good place to ask. Improvisers already have conventions for listening to, building on and refusing one another's offers, and improvisation has been argued to be a research method grounded in risk, active listening and collaborative response (Caines, 2020). A system that transforms and labels offers enters those conventions, and may quietly change them.

2 Related work

Existing research examines human–AI co-creation, collective decision-making and the attribution of creative credit; this project investigates their intersection in small-group digital improvisation. Table 1 compares the closest studies.

Table 1: Closest related work and the proposed extension, as reported in each source.

Study	Setting, method, duration	AI role	Focus and relevant finding	Extension in this project
Xu et al. (2026)	18 students in six groups of three, one ∼45-min design task, observed; 9 separate interviews	Implemented: one shared ChatGPT interface	Information flow, roles, negotiation; AI output became a negotiable part of decisions. Authors note trust and accountability received less attention	Improvisation; one person's transformed material; credit across sessions
Trajkova et al. (2024)	24 dance students; focus groups on partnered improvisation	Proposed AI dance partner; recommendations not yet tested	How improvisers lead, follow, introduce and value ideas	An implemented mediator in group work
Mathewson and Mirowski (2017)	Live improvised theatre; development report	Implemented chatbot performers	Successes and failures of performing alongside AI	AI transforming other people's offers
Epstein et al. (2020)	Online vignette experiments	Described, not used	Framing AI as an agent shifted responsibility away from the artist	Credit claimed in situ by the people making the work
Caines (2026)	Research-creation study of the author's course design	Web GenAI tools in teaching	Ethics, access, improvisatory skills	Contribution-level recognition in group improvisation
Proposed study (planned)	Three groups, four workshops each; episode analysis, interviews	Language-model mediator; human-mediated contrast	Questions: how recognition, authority and authorship form	—

Notably, Xu et al. (2026) coded "unacknowledged contributions" as one kind of breakdown. Contestability, developed mainly for high-stakes decisions such as hiring or sentencing (Lyons et al., 2021), offers a related vocabulary for challenging a system.

Communication scholarship offers two vocabularies. Human–machine communication asks how people make sense of AI as a communicator in its own right (Guzman and Lewis, 2020); AI-mediated communication asks what happens when AI alters messages between people (Hancock et al., 2020). This project draws on both, since participants may treat the system as a partner in one moment and as a channel distorting a colleague's offer in the next. Cultural scholarship explains why recognition matters. Jenkins et al. (2006, p. 3) define a participatory culture as one in which "members believe their contributions matter, and feel some degree of social connection with one another"; this study asks what happens to that belief when a model, rather than a peer, first responds to a contribution. Chalk (2022) treats Actual Play, improvised tabletop play performed as media, as cultural production in which few producers earn a living and non-economic values

such as participation, enjoyment and community help motivate the work. Recognition is thus part of what participants receive for their labour. Neville (2026) shows how technologies are rolled out into settings whose practices outsiders misrecognise as non-technical; improvisers' conventions for crediting and reusing offers are such practices, and my system's attribution rules could override them. I take from these works the stakes of recognition, and question whether a system's record of contribution can stand in for it.

3 Key terms and research questions

Three working definitions organise the study. **Recognition**: a contribution is noticed or acknowledged as meaningful. **Creative authority**: a person's interpretation or decision carries influence over what happens next. **Authorship**: participants associate ownership, credit or responsibility with the work, as felt ownership, public attribution and creative responsibility. Legal ownership and copyright are outside the scope of this study. These can come apart: a contributor may be credited while the group reshapes the material regardless of their view (credit without influence), and a facilitator, interpreter or I as designer may shape the work uncredited (influence without credit). The definitions are sensitising concepts, not fixed categories, and listening, facilitation, silence and refusal count as participation.

Central question. In collaborative digital improvisation, how do participants, facilitators and a generative-AI interface shape whether a person's contribution is recognised, carries creative authority, and is credited in the shared work?

RQ1. *Recognition.* How do participants notice, interpret and respond to differences between what they offered and how it was transformed?

RQ2. *Authority.* Through which practices, including acceptance, revision, refusal, facilitation and silence, do contributors and groups gain or lose influence over transformed material, and how do interface permissions shape those practices?

RQ3. *Authorship.* When and why do the system's technical attribution and participants' accounts of ownership, credit and responsibility diverge?

4 Theoretical framework

My primary framework is Lucy Suchman's study of human–machine reconfiguration (Suchman, 2007), which examines how agencies are figured at the interface and how humans and machines come to be treated as similar or different in practice. On this view, whether a transformed element belongs to its contributor, to the group or to the system is worked out in interaction.

In the illustrative episode in Section 5, the system records that an amber pulse came from one participant's phrase, but the pulse also depends on the model's reading, my asset library and the approval rule. When the contributor refuses a request to brighten it, they enact an authority the interface granted, which the group can

accept or contest. The question shifts from "who really made the pulse?" to how agency is distributed and attributed in that moment: how participants attribute agency in talk and action ("the AI did that", "that one is mine"), how rules, model defaults, facilitation and my design decisions make some actions easy and others difficult, and who is held responsible when a transformation misrepresents someone. Improvisation studies adds a limited lens on how offers are taken up, ignored or refused (Caines, 2020). These concepts tell me where to look, not what I will find.

5 Research-creation methodology

The project follows the program's research-creation option, a creative work with a 150–200-page written dissertation (Joint Graduate Program in Communication & Culture, 2021). It is closest to what Chapman and Sawchuk (2012) call "research-from-creation": making the work generates material for understanding how participants engage with it.

The creative work. The creative component will comprise the shared audiovisual environment and the improvised works produced through it. Developing and revising its expressive possibilities, interaction rules and workshop scores will form part of the inquiry into how creative authority is distributed. Participants build and perform short projected improvisations of light, form and sound, then discuss them; each rule the environment embodies is a proposition about collective authorship that they can test, resist or rewrite. In an illustrative session, four improvisers receive a score: build the weather of a city none of them has visited. One types "the bus that never comes." The system proposes a slow amber pulse at the edge of the space with a low drone, with the generated interpretation "waiting as a distant repeating light." The contributor accepts it. Later another participant wants the pulse brighter; the contributor declines, and the facilitator asks the group how to proceed. The scores are a creative output in their own right: a revised set of improvisation scores for AI-mediated group work, credited to the groups that shaped them with their consent, which other practitioners can use, adapt or refuse. Because the creative work must be available for the committee to assess (Joint Graduate Program in Communication & Culture, 2021), it will be documented through the executable environment, these scores, selected consented recordings, and an annotated account of design revisions. These materials will allow the committee to examine how the creative process contributes to the dissertation's argument.

Ownership and reuse. Recording origins and giving contributors approval rights could introduce individual ownership where improvisers treat accepted offers as shared. Scoping interviews will investigate how practitioners already understand ownership and reuse of improvised offers, including whether accepted material becomes the group's and who decides that. Co-design will examine whether contributor approval and visible attribution support their practices or introduce individualised ownership assumptions that participants do not share. One option to test is a way to "release" accepted material to the group.

Phases and provisional scale. Scoping interviews with improvisers and facilitators come first. Co-design sessions with a few practitioners then set the input form, the limits of transformation, the starting approval and attribution rules, and accessibility provisions; these practitioners will not join the workshop groups.

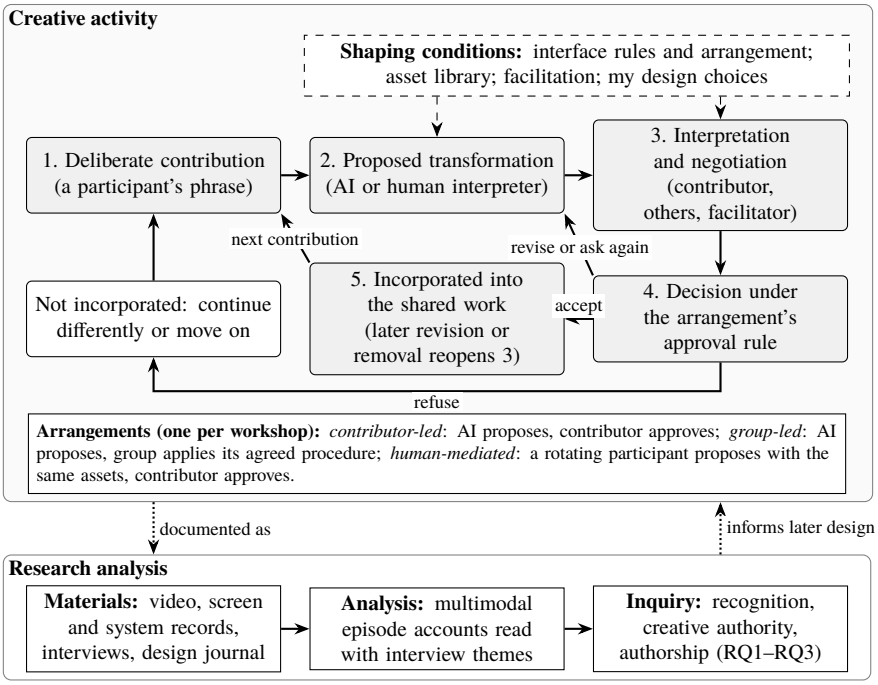


Figure 1: Proposed study (planned, not yet conducted). Solid arrows show a workshop's creative cycle and its decisions; dashed arrows show conditions shaping transformation and negotiation; dotted arrows link activity and analysis. Recognition and authorship are interpreted from the materials, not produced by system events.

A provisional design would then involve three groups of three to five adult practitioners, each attending four workshops of approximately 90–120 minutes, with recruitment seeking variation in improvisational and creative-technology experience. I would aim to include one existing ensemble and record prior relationships. Each participant then has a reflection interview reviewing video of their own sessions. The final design would be refined through scoping, access negotiations and ethics review.

Arrangements and what they support. The first workshop uses only the manual path, and the group writes down the procedure it will use when the group decides. Each later workshop uses one arrangement from Figure 1, in a different order for each group, so each group meets each arrangement once. What repeats is the group's evolving collaboration across sessions, not trials within an arrangement. The arrangements are situated contrasts whose effects cannot be separated from order, prior relationships and emerging norms; I treat these as contextual evidence. The human-mediated arrangement keeps contributor approval but also changes timing and social relationships, so the study will not isolate a causal effect of AI.

Negotiating the rules. The co-designed starting rules define each arrangement. During a session, participants may negotiate how they apply them, for instance by releasing material or overruling an approval; substantial system changes are made only between sessions. Each rule version, its reason and its consequences are recorded in the design journal. A changed arrangement is not treated as equivalent to its earlier version, and the change itself is analysed as evidence of negotiated authority.

6 The system and the place of AI

A participant submits a short text phrase. An existing language model returns one *proposal* drawn from a curated library of forms, light, colour, particle "weather", sound and text, with a one-line *generated interpretation* that is model output, not an explanation of why the model produced the transformation. Contributors or groups can accept, edit, reject, request another proposal or undo, and a manual path to the same library is always open. Generative AI is included because machine-mediated reinterpretation is the object of study; the project does not presume that AI improves improvisation. The model does not decide whether anything enters the work, who is credited or what is removed, and makes no inferences about participants' emotions or behaviour.

Neither the model nor the library is a neutral container. Documented challenges include stereotyped images from text-to-image systems (Bianchi et al., 2023), value profiles in five GPT models resembling English-speaking and Protestant European survey responses (Tao et al., 2024), and a preference for Western-associated entities among multilingual and Arabic models prompted in Arabic (Naous et al., 2024). A 2022 benchmark also found models of that time below human accuracy on paired figurative phrases (Liu et al., 2022). These results concern specific models and tasks, and I will not benchmark models. I will record how metaphor, idiom and culturally specific phrases are read when participants use them, discuss misreadings with them, and examine how my asset choices privilege particular associations.

Feasibility. Unity applies only permitted, bounded actions; model output is never run as code. I will first test a local open-weight model, otherwise an approved hosted service receiving only the phrase. No model is trained.

7 Case-study context and access

Subject to discussion with Dr. Caines, these workshops could examine how improvisation helps participants question and reshape the authority built into creative technologies, a question close to the title of her SSHRC Insight project, *Interfacing, Playtesting, Resetting: Reframing Technology Use Through Improvised Creative Technologies* (Goodfellow Craig, 2026). Any relationship to that project would be determined together; I do not assume it includes this study or any AI component. Otherwise I would use open calls to Toronto improvisation, theatre and sound communities and to AMPD students, excluding anyone I teach. As in Chalk's (2022) study, recruitment would be iterative, checking whose practices are missing after the first group and reporting remaining gaps. I will not work with disability-focused or Indigenous communities without established partnerships, community governance and relevant expertise.

8 Analysis

I will construct multimodal accounts of selected episodes, linking video of participants' words and actions with screen recordings and the system's log. An episode runs from a contribution, through its transformation and the group's response, to the next creative action. Episode selection will include routine acceptance and non-response as well as explicit disagreement. I will document selection decisions and compare episodes across groups, arrangements and stages of collaboration, including cases that challenge emerging interpretations. The episode analysis adapts the video microethnography McArthur (2015) used to study interface interactions; reflection interviews are analysed through reflexive thematic analysis (Braun and Clarke, 2019), whose themes contextualise an episode without settling what it meant. System records show what happened in the software, not intention. I will keep an analytic trail rather than seek inter-rater agreement, and the design journal records how my choices as designer and facilitator shape what participants can do.

9 Ethics, consent and accessibility

Consent will be specific, ongoing and revisable (Im et al., 2021; Strengers et al., 2021), given separately for recording, AI processing, use of recordings in the examination, and public dissemination; public sharing is optional. Provenance will support identifying and, where feasible, removing attributable contributions. Participants will agree on withdrawal procedures and deadlines before the workshops, including the limits of removing contributions from jointly developed or publicly disseminated works. Credit is agreed with participants, whose time is paid; data are minimised, pseudonymised and kept for stated periods. Full participation without AI is a design commitment the workshops will test, through reduced motion, resizable text, alternatives to sound-only information, facilitator-operated controls and the manual path. Participation alone does not make a project equitable (Birhane et al., 2022).

10 Intended contribution

The project investigates whether technical attribution and social recognition can diverge in collective creative work. A provenance log may record who supplied a phrase while collaborators treat the transformed material as another person's, the group's, or nobody's. The study will examine how contributors gain or lose authority over transformed material, and how interface permissions, facilitation and group conventions shape that process. Its intended contribution is an explanation of these relationships, rather than a general claim that more individual control produces fairer collaboration. The evidence may help explain credit without influence, influence without credit, how accepted material becomes collective, and when provenance fails to resolve authorship disputes; these are possibilities, not predicted findings. For communication studies, the project would extend AI-mediated communication from messages between individuals to collective creative production, where the mediated "message" becomes shared work. For improvisation practice, it would offer scores, tested in the workshops, for working with, around and against AI mediation. The resulting concepts

may help interpret other platforms that rely on attribution records, without establishing how those platforms behave. If AI proves unnecessary in some arrangements, the contribution would lie in explaining when and why.

11 Relationship to my master's research

My master's research at Simon Fraser University asks how interfaces make a person's state perceptible to others, through label-free heart-rate cues for remote teammates in a cooperative Unity game and a Unity environment that responds to the player's heart rate. Both manuscripts are under revision following peer review (Jahandideh, 2026a,b). In the teammate study, participants tied their willingness to use the cue to control over it, asking for opt-in sharing, an immediate off switch and separate permissions (Jahandideh, 2026b). As the designer, I also noticed that the classification behind the display still sorted people into evaluative categories they never saw. Both lead to this PhD's question, who controls how a shared system interprets people, now asked of deliberate creative contributions rather than bodily signals.

12 Provisional chapter structure

(1) Introduction; (2) literature and framework; (3) methodology: research-creation, workshops and reflexivity; (4) the environment as a creative work and the ownership its design assumes; (5) recognition and authority; (6) authorship, attribution and responsibility; (7) conclusion.

13 Provisional timeline (full-time)

Year 1 (2027–28)	Courses (CMCT 7000, 7200, 7005 in the current degree requirements, and electives; offerings may change); literature review; ethics preparation; tests with a local model.
Year 2 (2028–29)	Comprehensive examinations; formal proposal (maximum 3,500 words) with a supervisory committee (Joint Graduate Program in Communication & Culture, 2016); ethics approval; scoping interviews.
Year 3 (2029–30)	Co-design; workshops with three groups; reflection interviews; analysis.
Year 4 (2030–31)	Analysis; documentation of the creative work; writing and defence. Year 5 in reserve.

14 Program and supervisor fit

The project belongs to Technology in Practice, with Media & Culture as a minor. Dr. Caines's work on improvisation as a research methodology (Caines, 2020) and socially responsive creative technologies (Caines and Watson, 2023) speaks to its setting and methods. In her research-creation study of generative AI in classrooms, she came to see her work with AI tools "as a kind of critical improvisation practice," one involving

"a deep responsibility to how this material is utilized" and "a constant cycle of navigating power, adaptation, surprise, failure, and creation" (Caines, 2026, p. 193). That study centred on course design; this project asks how that responsibility and power are negotiated among improvisers when the material is one another's contributions.

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